SujetStore

Educational Archive Platform for the Algerian Curriculum

Technical Manual — Architecture, Features & Roadmap

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Table of Contents

1. Introduction: Rethinking Digital Learning in Algeria 4
    1.1. The Problem 4
    1.2. The Vision 4
    1.3. Core Objectives 4
2. System Architecture & Tech Stack 5
    2.1. The Tech Stack 5
    2.2. Architecture Philosophy: The Compact Monolith 5
    2.3. Dual-Database Strategy 5
        2.3.1. Content Database (`content.db`) — Read-Intensive 6
        2.3.2. Users Database (`users.db`) — Read-Write 6
    2.4. Development Workflow 6
3. UI/UX & Design System 6
    3.1. Native RTL & Arabic Typography 6
    3.2. Dark & Light Mode with Shadcn-Svelte Theming 7
    3.3. Mobile-First Responsive Design 7
    3.4. Global Navigation Hierarchy 8
4. The Document Engine 8
    4.1. Discovery & Categorization 8
    4.2. Trimester Tabs: Segmented Control 8
    4.3. The Contextual Filter Bar 8
    4.4. Smart PDF Viewer & Printing 9
    4.5. Smart Tracking Persistence 9
5. The Interactive Assessment Engine 9
    5.1. Architecture of the Quiz Engine 9
    5.2. The 14 Supported Question Types 9
        5.2.1. Standard Selections 10
        5.2.2. Drag & Drop Interactions 10
        5.2.3. Text Inputs 10
        5.2.4. Visual & Touch Interactions 10
    5.3. Instant Feedback & Analytics 10

6. Gamification & Engagement 11
6.1. Points System 11
6.2. Achievement Badges 11
6.3. The Achievements Dashboard 11
7. Administrative Dashboard & Content Management 12
7.1. Security & Authentication 12
7.2. The Document CMS 12
7.3. The Interactive Quiz Builder 12
7.4. Analytics Dashboard 13
8. Deployment & Future Roadmap 13
8.1. Production Deployment Guide 13
8.1.1. Current Infrastructure Footprint 13
8.1.2. Deployment Commands 13
8.2. Security Layers 13
8.3. Future Roadmap 14
8.3.1. Phase 2: Video Solutions & Streaming Architecture 14
8.3.2. Phase 3: Advanced Analytics & Weakness Detection 14

1. Introduction: Rethinking Digital Learning in Algeria

1.1. The Problem

In the current Algerian educational landscape, students face a fragmented digital experience. Exam papers, quizzes, and study materials are scattered across social media groups, unmoderated forums, and low-quality scan repositories. Key issues include:

- **Content Dispersion:** Thousands of documents are posted daily across Facebook groups and Telegram channels with no indexing, versioning, or quality control. Finding a specific exam from a specific trimester becomes an exercise in frustration.
- **No Pedagogical Structure:** Materials are dumped without classification by trimester, type, difficulty, or alignment with the official Ministry of Education curriculum.
- **Poor Visual Quality:** Most documents are low-resolution phone scans with inconsistent fonts, skewed pages, and illegible handwriting.
- **Hostile User Experience:** Existing platforms rely on aggressive advertising, intrusive pop-ups, and confusing navigation that actively hinders the learning process.

1.2. The Vision

SujetStore was conceived as a purpose-built, open-source educational archive platform that directly addresses these pain points. It provides a clean, focused, and fast experience for browsing, filtering, and studying exam papers, interactive quizzes, and study materials for the Algerian curriculum — from primary school through high school.

1.3. Core Objectives

- **Absolute Focus:** Deliver content within a distraction-free, ad-free interface designed for deep study and review. No pop-ups, no banners, no autoplay videos.
- **Hierarchical Archiving:** Organize the entire curriculum into a strict, navigable taxonomy: *Level* $\longrightarrow$ *Year* $\longrightarrow$ *Subject* $\longrightarrow$ *Trimester* $\longrightarrow$ *Document Type*.
- **Interactive Learning:** Move beyond static PDFs by providing a robust 14-type quiz engine that tests comprehension instantly.
- **Empowering Content Creators:** Provide a powerful yet simple admin CMS that allows educators to upload, categorize, and manage documents and interactive quizzes effortlessly.
- **Resilient Infrastructure:** Leverage a modern, lightweight stack (SvelteKit + Bun + SQLite) that can scale to millions of concurrent users without infrastructure collapse.

2. System Architecture & Tech Stack

SujetStore is built on a modern, performance-oriented stack carefully selected for speed, simplicity, and developer experience.

2.1. The Tech Stack

Component	Technology	Rationale
Runtime	Bun (v1.1+)	3× faster than Node.js. Built-in bundler, test runner, and native SQLite driver.
Framework	SvelteKit (Svelte 5)	Full-stack framework with SSR. Powerful reactive runes (`$state`, `$derived`). Minimal client-side overhead.
Database	SQLite	Embedded, zero-config, read-optimized. Uses `bun:sqlite` for native performance.
ORM	Drizzle ORM	Lightweight, type-safe query builder with full TypeScript support.
CSS	TailwindCSS v4	Utility-first CSS with native CSS variable support and zero-config setup.
UI Components	Shadcn-Svelte	Accessible, themeable component library with semantic color tokens.
Auth	Custom (bcryptjs)	Session-based authentication stored in SQLite. No external dependencies.
Deployment	Caddy + systemd	Auto-SSL reverse proxy with process management.

2.2. Architecture Philosophy: The Compact Monolith

Rather than adopting microservices or a headless frontend/backend split, SujetStore uses a **Compact Monolith** architecture:

- **Single Process:** The frontend (Svelte 5) and backend (SvelteKit server routes) run within a single Bun process.
- **File-Based Routing:** Routes are defined by the filesystem structure under `src/routes/`, eliminating the need for manual route configuration.
- **Co-located Data:** Both SQLite databases (`content.db` and `users.db`) live on the same server, accessed via `bun:sqlite` for sub-millisecond query performance.

2.3. Dual-Database Strategy

The system uses two separate SQLite databases to cleanly separate concerns:

2.3.1. Content Database (`content.db`) — Read-Intensive

Stores all educational content: levels, years, subjects, trimesters, documents, quizzes, and quiz questions. This database is generated offline using the seed script and deployed as a single file. Updates are often performed by replacing the entire file or making bulk inserts through the Admin panel.

2.3.2. Users Database (`users.db`) — Read-Write

Stores admin user accounts, session data, user gamification data (points, badges), quiz attempts, ratings, comments, and global notifications. WAL mode is enabled for concurrent read/write access:

```
PRAGMA journal_mode = WAL;
PRAGMA busy_timeout = 5000;
PRAGMA synchronous = NORMAL;
```

This ensures that high-volume writes (like tracking points and attempts) never block reads on the content database.

2.4. Development Workflow

```
# Install dependencies using Bun
bun install

# Seed the databases with the Algerian default tree (Primary, Middle,
Secondary)
bun run scripts/seed-education.ts
bun run scripts/seed-algerian-questions.ts

# Start the development server
bun run dev

# Type-check the project
bunx svelte-check
```

3. UI/UX & Design System

SujetStore was designed with the principle that *"simplicity creates effectiveness."* The interface feels native, fast, and accessible regardless of the device.

3.1. Native RTL & Arabic Typography

The entire design is built from the ground up for Right-to-Left (RTL) text direction.

- **Typography:** The `Cairo` font from Google Fonts is applied globally. It was selected for its exceptional legibility in Arabic numerals and varied stroke weights, making it ideal for both long reading (like document summaries) and bold headers.
- **Layout Engine:** Flexbox and CSS Grid are utilized exclusively with logical CSS properties (`padding-inline`, `margin-inline-start`) to ensure perfect mirroring natively without any hacky overrides.

3.2. Dark & Light Mode with Shadcn-Svelte Theming

The platform implements a complete dual-theme system using `Shadcn-Svelte` semantic color tokens. Every component — from buttons and dropdowns to cards and toggle switches — automatically adapts to the user's preferred color scheme based on their system setting, with a manual override toggle available in the navigation bar.

Key semantic tokens used throughout the interface:

- `bg-background` / `text-foreground` — Base page colors
- `bg-card` / `text-card-foreground` — Elevated surface containers (like Document boxes)
- `bg-muted` / `text-muted-foreground` — Subdued backgrounds and secondary helper text
- `bg-primary` / `text-primary-foreground` — Active elements, CTAs, and primary buttons
- `bg-accent` / `text-accent-foreground` — Hover and interactive states

This approach guarantees that **no hardcoded hex colors** appear in the core UI codebase (excluding specific gamification badges), ensuring perfect visual contrast in both modes.

3.3. Mobile-First Responsive Design

Recognizing that the vast majority of Algerian students access educational content from smartphones, the interface implements strict mobile-first paradigms:

- **Stacked Controls:** Filter controls stack vertically on small screens and align horizontally on desktop displays.
- **Horizontal Scrolling:** Tab navigation (for trimesters and quiz stats) scrolls horizontally with hidden scrollbars when screen width is constrained to preserve vertical screen real estate.
- **Touch Targets:** All interactive elements (buttons, checkboxes, drag-and-drop handles) meet or exceed the minimum touch target size of 44×44px required by accessibility guidelines.

3.4. Global Navigation Hierarchy

A global breadcrumb navigation system (`Level / Year / Subject`) is present on all inner pages. This ensures the student always knows their exact location in the curriculum taxonomy and can navigate back to a parent category with a single click, completely eliminating the feeling of being "lost" inside the application.

4. The Document Engine

The core utility of SujetStore is the rapid discovery, filtering, and consumption of static educational documents (Exams, Assignments, Summaries).

4.1. Discovery & Categorization

The homepage presents the three educational levels of the Algerian system:

- **Primary** — Grades 1 through 5
- **Middle School** — Grades 1AM through 4AM
- **Secondary** — Grades 1AS through 3AS (across all specialized streams)

Each drill-down reveals a structured grid of years, leading to the core subjects.

4.2. Trimester Tabs: Segmented Control

Inside a subject page, documents are grouped using a **Segmented Control** interface — a horizontal tab bar with three segments: *Term 1*, *Term 2*, *Term 3*, and *General*.

Built on Shadcn UI's `Tabs` pattern, this control physically separates the academic year. Furthermore, the system implements **Smart Auto-Tab Selection**: On page mount, the engine scans the trimesters in order and automatically activates the first one that contains available documents, saving the user from clicking through empty tabs.

4.3. The Contextual Filter Bar

Inside each trimester tab, users can narrow down hundreds of documents instantly using client-side Svelte 5 `$derived` runes:

- **Document Type:** Filter by Exam, Assignment, Lesson, Summary, or Book.
- **Source:** Filter by Official (Ministry standard) or Proposed (Teacher drafts).
- **Strict Solution Toggle:** A prominent switch to show "Only documents with solutions attached."

Because filtering happens entirely client-side, the UI updates instantly without any network latency or skeleton loaders.

4.4. Smart PDF Viewer & Printing

SujetStore integrates an in-browser PDF viewing experience that keeps students on the platform without forcing downloads:

- **Fullscreen Modal:** Clicking a document opens a distraction-free, full-height modal containing the PDF `iframe`.
- **One-Click Solution Toggle:** If a document possesses an attached solution file, a floating toggle button allows the student to flip instantly between viewing the "Subject" and the "Correction" without reloading the modal.
- **Background Printing:** Print functionality is handled seamlessly, opening the system print dialog directly from the file blob to ensure perfect margins regardless of browser quirks.

4.5. Smart Tracking Persistence

To help students track what they have studied over long periods, the application implements local persistence tracking. When a student opens a document, its ID is saved to the browser's `localStorage` via a `Set` implementation.

When displaying the document lists, previously viewed items are marked with a distinct Checkmark indicator (✓) and a slightly faded styling, providing immediate visual feedback on their revision progress.

5. The Interactive Assessment Engine

To complement the static document archive, SujetStore features a fully functional, interactive quiz engine that allows students to test their knowledge and receive instant feedback.

5.1. Architecture of the Quiz Engine

The assessment engine is deeply integrated with the platform's core architecture:

- **JSONB Storage Approach:** All quiz questions are stored in a single `questions` table. Instead of creating separate database tables for every question type (the approach used by older systems like Moodle), SujetStore uses a single `question_data` text column. This column stores schema-validated JSON payloads specific to each question type, offering massive flexibility for adding new types without running database migrations.
- **Dual Modes:** Students can take a quiz in "Practice Mode" (instant feedback per question) or "Exam Mode" (timed, all answers submitted at once).

5.2. The 14 Supported Question Types

The platform supports a comprehensive set of 14 interactive Moodle-style question types, categorized as follows:

5.2.1. Standard Selections

- **MCQ:** Multiple choice questions with single or multiple correct answers.
- **True/False:** Binary choice with visual action buttons.
- **Matrix:** A multi-row grid selection (e.g., True/False evaluation per statement row).

5.2.2. Drag & Drop Interactions

- **Ordering:** Native drag-and-drop handles to reorder items into a correct sequence.
- **Drag & Drop:** Categorize items by dragging them into corresponding drop zones.
- **Matching:** Connect pairs of related terms from two separate columns.

5.2.3. Text Inputs

- **Fill in the Blank:** Inline text inputs replacing gap placeholders within a paragraph.
- **Short Answer:** Free text input with keyword matching and tolerance validation.
- **Cloze:** Inline gaps with dropdown selection menus instead of free text.
- **Essay:** Free-text area with word limit tracking and optional keyword-based auto-grading.
- **Calculated:** Formula-based questions (e.g., `x + y = ?`) where variables are randomized per student, and answers are evaluated against a tolerance margin.

5.2.4. Visual & Touch Interactions

- **Sentence Reorder:** Tap words to reassemble them into a grammatically correct sentence (optimized for mobile).
- **Hotspot:** Click or tap on the correct defined area of an uploaded image.
- **Drag to Image:** Place floating text labels onto a background diagram at their correct geometric positions.

5.3. Instant Feedback & Analytics

When a student submits an answer, the evaluation logic runs server-side to prevent cheating. The UI immediately displays:

- A clear visual indicator (Red/Green or Cross/Checkmark).
- The correct answer (if the student was wrong).
- An optional explanation or hint defined by the teacher.

Results are saved to the `user_points` and `quiz_attempts` tables to power the Gamification system.

6. Gamification & Engagement

To increase student engagement, motivation, and retention, SujetStore implements a complete gamification layer. This transforms the solitary act of studying into a rewarding, trackable journey.

6.1. Points System

The foundation of the gamification system is the Points ledger. Points are awarded automatically for positive educational behaviors:

- **Completing a Quiz:** +10 points.
- **Achieving a Perfect Score:** Bonus points depending on quiz length.
- **Earning a Badge:** Bonus points ranging from +10 to +50 based on difficulty.

All point transactions are recorded in the `user_points` table with a timestamp and reason, providing students with a transparent history of their earnings.

6.2. Achievement Badges

The platform defines 8 built-in achievement badges that students unlock by reaching specific milestones in the Quiz Engine.

Badge	Icon	Unlock Condition
First Quiz	🎯	Complete your very first interactive quiz
Quiz Explorer	📚	Complete 5 total quizzes
Quiz Master	🏆	Complete 10 total quizzes
Perfect Score	⭐	Score 100% on any quiz
Speed Demon	⚡	Complete a quiz in under 1 minute
Hot Streak	🔥	Score 80%+ on 3 consecutive quizzes
Versatile	🎪	Try 5 different interactive question types
Centurion	💯	Accumulate 100 total points

The badge evaluation logic runs asynchronously upon quiz submission. When a new threshold is met, the system inserts a record into the `user_badges` table.

6.3. The Achievements Dashboard

Students can view their progress on the dedicated Achievements page (`/achievements`). This page utilizes a horizontal tab layout to present three key views:

1. **Badges Grid:** A visual display of all 8 badges. Unearned badges appear grayscale/locked, while earned badges are fully colored with their unlock timestamps.

2. **Leaderboard:** A global ranking showing the top 10 most active students on the platform, ranked by total accumulated points.
3. **Points History:** A chronological log of the student's recent point earnings and the specific actions that triggered them.

7. Administrative Dashboard & Content Management

A platform is only as good as its content. SujetStore provides administrators and educators with a powerful, custom-built Content Management System (CMS) to manage the entire educational tree without touching the database.

7.1. Security & Authentication

Access to the `/admin` routes is strictly protected.

- Authentication uses a custom implementation relying on `bcryptjs` for password hashing.
- Sessions are stored in the `users.db` SQLite database via secure HTTP-only, `SameSite=Lax` cookies.
- Rate limiting is enforced on login endpoints to prevent brute-force attacks.

7.2. The Document CMS

The administration panel provides full CRUD (Create, Read, Update, Delete) capabilities for the taxonomic hierarchy:

1. **Levels** (e.g., Middle School)
2. **Years** (e.g., 4AM)
3. **Subjects** (e.g., Mathematics)
4. **Documents** (e.g., Term 1 Exam with Solution)

Key UI Features:

- **Modal-Based Editing:** All add/edit operations use floating Shadcn-Svelte modals for fast interaction without triggering full page reloads.
- **Confirmation Dialogs:** Destructive operations require explicit confirmation through a `ConfirmModal` component to prevent accidental data loss.
- **File Linking:** Documents can be linked directly via direct URL, ensuring the platform database remains lightweight.

7.3. The Interactive Quiz Builder

Creating interactive Moodle-style content is notoriously complex in most LMS platforms. SujetStore solves this by providing a highly optimized, visual Quiz Builder.

Builder Features:

- **Drag-and-Drop Ordering:** Teachers can reorder questions within a quiz visually.
- **Specialized Question Sub-forms:** Because the platform supports 14 different question types, the builder dynamically renders a tailored sub-form based on the selected type.
  - For example, selecting "Matrix" reveals a grid builder for rows and columns.
  - Selecting "Hotspot" reveals an image upload field and X/Y coordinate inputs for defining clickable zones.
- **Quiz Metadata Engine:** Set time limits, required passing scores, and publish toggles.

7.4. Analytics Dashboard

The admin home page provides a high-level overview of platform health:

- Total Document counts grouped by Level and Year.
- Global point accumulations.
- Quiz completion statistics to identify the most difficult or popular exercises across the platform.

8. Deployment & Future Roadmap

8.1. Production Deployment Guide

SujetStore is designed to be easily deployed on cheap VPS infrastructure (e.g., Contabo, Hetzner) while handling massive scale.

8.1.1. Current Infrastructure Footprint

Component	Specification
Provider	Standard Linux VPS (Ubuntu 22.04+)
Web Server	Caddy (auto-SSL reverse proxy)
Process Manager	systemd service unit
Backup Strategy	Daily SQLite `.backup` of `users.db` with 7-day retention

8.1.2. Deployment Commands

A deployment script (`deploy.sh`) is provided in the repository root to automate the build and restart process.

```
./deploy.sh full    # Initial setup, dependency install + deployment
./deploy.sh deploy  # Subsequent updates (git pull, build, restart)
```

8.2. Security Layers

Layer	Implementation
Transport	HTTPS enforced via Caddy auto-SSL
Cookies	`HttpOnly` · `Secure` · `SameSite=Lax`

Passwords	bcryptjs hashing for Admin accounts
SQL Injection	Guaranteed prevention via Drizzle ORM prepared statements

8.3. Future Roadmap

While Phase 1 (The Interactive Assessment Engine) is now successfully completed, SujetStore has clear expansion vectors for Phase 2 and beyond.

8.3.1. Phase 2: Video Solutions & Streaming Architecture

To support deeper understanding—especially for subjects like Physics and Mathematics—the platform will integrate video walkthroughs mimicking a teacher's explanation.

Proposed Streaming Tech Stack:

- **HLS Protocol:** Adaptive bitrate streaming with 10-second chunks for instant playback.
- **AES-128 Encryption:** Per-chunk encryption to prevent unauthorized downloads.
- **Backblaze B2:** Cold storage at a fraction of the cost of Amazon S3.
- **Cloudflare CDN:** Zero egress fees via the Bandwidth Alliance, providing rapid edge distribution across North Africa.

8.3.2. Phase 3: Advanced Analytics & Weakness Detection

Transitioning from macro-level admin analytics to personalized student diagnostics:

- **Automated Weakness Detection:** Analysis of specific interactive quiz errors to identify targeted knowledge gaps (e.g., "Struggles with Fractions").
- **Personalized Recommendations:** Automatically suggesting specific summaries or lessons based on failed quiz competencies.
- **Parent Dashboards:** Summary statistics tracking a student's study time, exam scores, and progression by trimester.